Universal Radiative Corrections for Precision Determinations of V_{ud}

Ana Costa Pereira, Martin Gorbahn, Francesco Moretti, Kai Sieja, Emmanuele Stamou

Ana.Costa-Pereira@liverpool.ac.uk

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1. Semi-leptonic kaon decays
2. Ratio of V_{us}/V_{ud}
3. Radiative corrections and EFT formalism
4. NLO EW corrections
5. NNLO EW corrections

- SM shortcomings: Origin of flavour, CP problem, dark energy, dark matter, matter/anti-matter asymmetry;
- CKM (mixing of quark flavour eigenstates) tested at level 0.01%;
- Cabbibo angle anomaly;
- Measure decay rates: K, $0^+ \longrightarrow 0^+$, Π, neutron decay;
- Semileptonic decays: K_{l3}, $0^+ \longleftarrow 0^+$, neutron decay;
- Leptonic decays: K_{l2};
- Status of V_{ud}, V_{us};

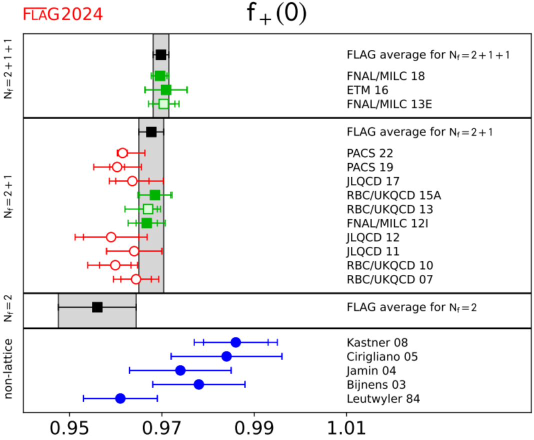
K_{l3} - Semi-leptonic kaon decays

$$\Gamma(K_{\ell 3}) = \frac{G_F^2 M_K^5}{192\pi^3} C_K^2 \, S_{\rm EW} \, |V_{us}|^2 \left| f_+^{K^0\pi^-}(0) \right|^2 I_{K\ell}^{(0)} \left(1 + \delta_{SU(2)}^{K\pi} + \delta_{\rm EM}^{K\ell}\right)^2$$

- $S_{EW} = 1.0232(3) \longrightarrow$ process indepented EW corrections;
- $I_{K\ell}^{(0)}$ phase space in isospin limit;
- $\delta_{SU(2)}^{K\pi}$ strong isospin breakig correction
- $\delta_{\rm EM}^{K\ell}$ long distance EW corrections

Experimental inputs

$$|V_{us}| \, f_+(0) = 0.21654(41)$$



$$f_+(0) = 0.9698(17)$$

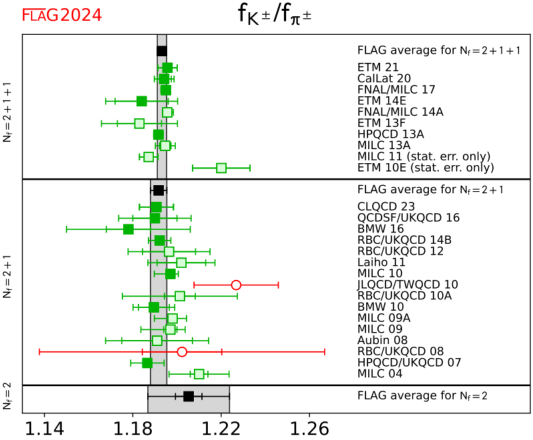
$$|V_{us}| = 0.22328(58) \qquad (K_{\ell 3})$$

$$\frac{\Gamma(K^{\pm}_{\mu 2(\gamma)})}{\Gamma(\pi^{\pm}_{\mu 2(\gamma)})} = \left|\frac{V_{us}}{V_{ud}}\right|^2 \left(\frac{f_{K^\pm}}{f_{\pi^\pm}}\right)^2 \frac{M_{K^\pm}\left(1 - m_\mu^2/M_{K^\pm}^2\right)^2}{M_{\pi^\pm}\left(1 - m_\mu^2/M_{\pi^\pm}^2\right)^2} \left(1 + \delta_{\mathrm{EM}}\right)$$

Experimental inputs

$$\left|\frac{V_{us}}{V_{ud}}\right| \frac{f_{K^\pm}}{f_{\pi^\pm}} = 0.27599(41)$$

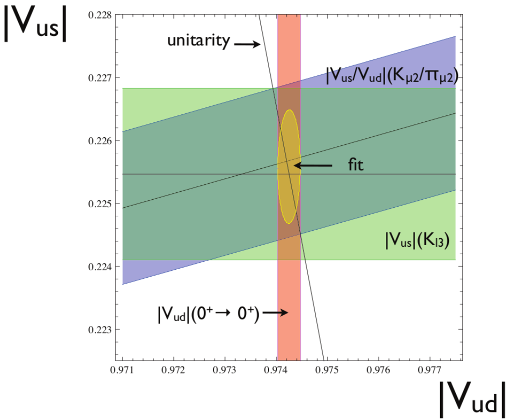
$$|V_{ud}| = 0.97373(31) \qquad \text{(superallowed } \beta \text{ decays)}$$



$$\frac{f_{K^\pm}}{f_{\pi^\pm}} = 1.1934(19)$$

$$\left|\frac{V_{us}}{V_{ud}}\right| = 0.23126(50) \qquad (K_{\ell 2}/\pi_{\ell 2})$$

CKM matrix



$$|V_{ud}|^2 + |V_{us}|^2 - O|V_{ub}|^2 = 1 \tag{1}$$

Operator product expansion

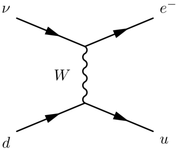


Figure 1: SM tree-level.

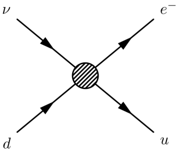


Figure 2: EFT Tree-level.

$$\frac{1}{k^2 - m_W^2} \longrightarrow \frac{-1}{m_W^2} \tag{2}$$

$$H_{eff}(x) = \frac{G_F}{\sqrt{2}} V_{CKM}^i C_i(\mu) Q_i(x) \tag{3}$$

NLO EW corrections

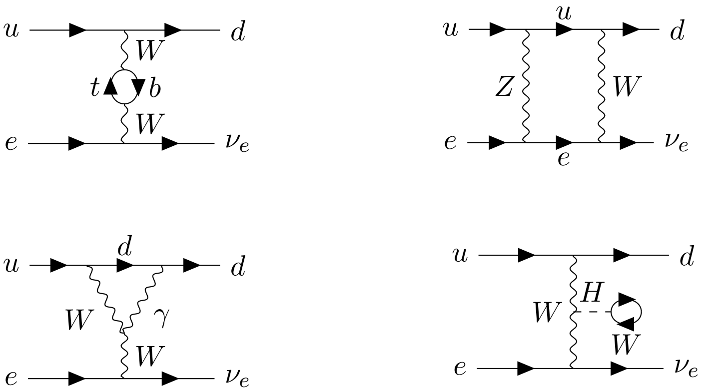


Figure 3: Some 1 loop diagrams in the full theory for the semi-leptonic decay.

Normalization to G_F

$$\mathcal{L}_{G_F} = -\frac{G_F}{\sqrt{2}} Q_\mu - \frac{G_F}{\sqrt{2}} V_{ud} C_q Q_q \tag{4}$$

And the Wilson Coefficients are

$$C_r^{LO} = 1 \tag{5}$$

$$C_r^{NLO} = -5 + \frac{a_\mu^1}{4} + \frac{a_q^1}{12} - 2\log\left(\frac{\mu^2}{M_Z^2}\right) . \tag{6}$$

- Finite, scheme dependent (a_μ^1, a_q^1)
- Only box diagrams contribute

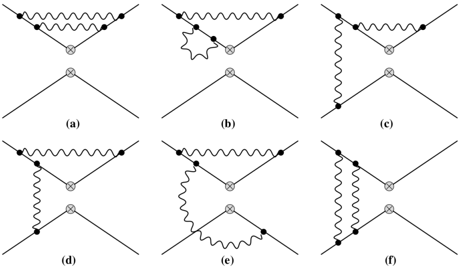


Figure 4: EFT NNLO EW corrections.

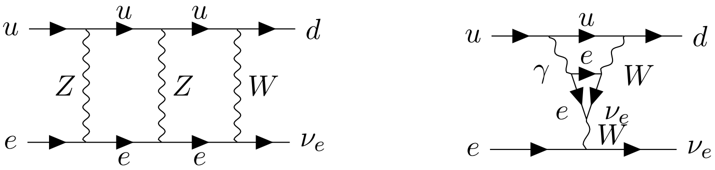


Figure 5: SM NNLO EW corrections.

WC and RGE

RGE for WC:

$$\mu \frac{d}{d\mu} C_i^f(\mu) = \sum_j \gamma_{ij}^f \, C_j^f(\mu), \tag{7}$$

$$C^{(f)}(\mu) = J_f(\mu)\, U_f(\mu)\, U_f^{-1}(\mu_0)\, J_f^{-1}(\mu_0)\, C^{(f)}(\mu_0) \tag{8}$$

insert graph with scale dependence

Conclusions

- Combination of lattice inputs and perturbative calculations;
- Higher order calculation + sub-percent lattice calculations;
- Current ongoing work on NNLO EW corrections for V_{ud} extraction from neutron decay.